Optimizing energy use in electric vehicles

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Introduction

Taking advantage of route features such as elevation in conjunction with regenerative braking promises to significantly reduce energy consumption of electric vehicles, reducing costs and potentially the number of charging stops. Conversely, driving efficiently yields a larger remaining range given the current state of charge. Using dynamic programming, it is possible to find an optimal choice of driving speed in order to minimize the energy necessary to cover a route in a certain time [**1**].

Methodology

The underlying model for the specific kinetic energy $e_{kin} = \frac{1}{2}v^2$ is given by

$$\frac{\mathrm{d}e_{kin}}{\mathrm{d}s} = A - C \cdot e_{kin}, \quad e_{kin}(0) = e_0 \ . \tag{1}$$

For the optimization, the following analytical solution is used:

$$e_{kin}(s) = \frac{A}{C} + \left(e_{kin}(0) - \frac{A}{C}\right) e^{-C \cdot s} \tag{2}$$

where $C = \rho_{air} \cdot C_D \cdot A_{front}/m$ describes air resistance (drag), while $A = \rho_{rot} \cdot M - g \cdot \tan\theta - g \cdot C_r$ includes the tractive force (acceleration and braking) applied by the drive system, the route resistance (especially the slope) and constant components of the travel resistance (rolling resistance) [**2**].

The route is split up into sections such that A and C are constant in each section. The resulting optimization problem is solved using dynamic programming with discrete values for the velocity and elapsed time at each section border.

Regenerative braking is taken into account by applying a discount factor (recuperation efficiency) wherever the tractive force is negative.

Setup

The optimized driving strategy (Fig. 1) is compared to a conventional one on a fictional route with different speed limits and slopes. The conventional driving strategy consists of a phase of constant acceleration with $2\frac{\mathrm{m}}{\mathrm{s}^2}$ until the current speed limit is reached, a phase of keeping a constant velocity and a phase of deceleration with $2\frac{\mathrm{m}}{\mathrm{s}^2}$ before reaching the next speed limit.

Different travel times are achieved by clipping the highest speed limits to lower values, corresponding to a voluntarily reduced maximum velocity. Both the optimized and the conventional driving strategy yield a tradeoff between the average velocity (or travel time) and the used energy, from which a desired compromise can be selected (Fig. 2).

Results

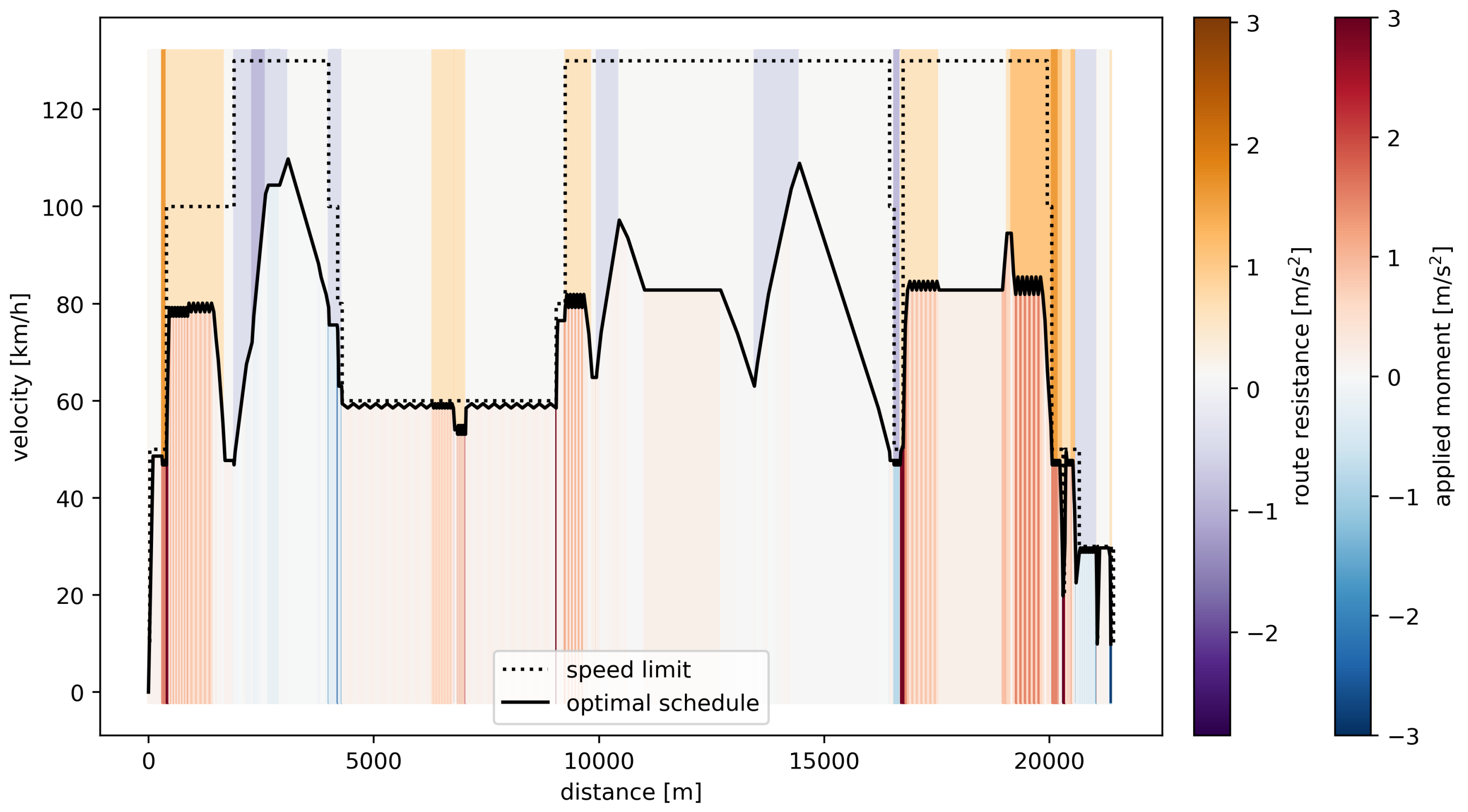

Figure 1: Optimal driving strategy (black line) for an average velocity of 64 km/h. Route resistance (slope and rolling resistance) is indicated by the coloring above the curve, the optimal traction force below. The dotted line shows the speed limit.

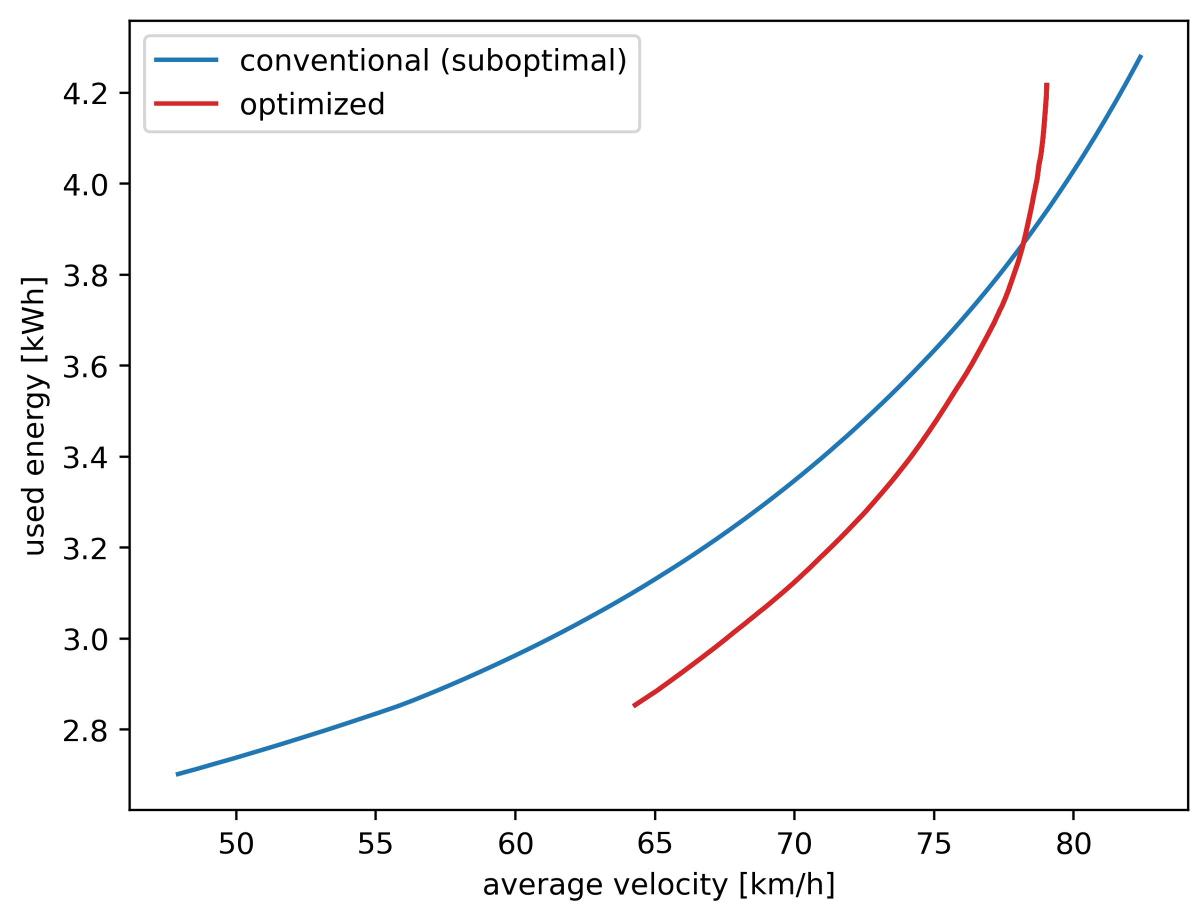

Figure 2: Tradeoff between average velocity and energy consumption. Comparison between conventional (blue) and optimized (red) driving strategies.

The optimized driving strategy results in a reduction of energy usage by about 7% compared to the conventional approach, while achieving the same average velocity.

Acknowledgements

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References

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[2] R. Franke, P. Terwiesch, M. Meyer, C. Klose, K.H. Ketteler, "Method for optimizing energy in the manner in which a vehicle or train is driven using kinetic energy." *US6668217B1, United States Patent Office, 23 December 2003.*